

3(a)	any four from: 1 identical to human, insulin / protein ; R similar 2 no, allergic reaction / immune response / side effects ; 3 human protein may, have high(er) activity / work better / (more) rapid response ; 4 no chance of developing tolerance (to animal insulin) ; 5 no risk of transmitting diseases ; 6 supply unlimited / large-scale production ; ora not enough animal pancreases to meet demand 7 cost (of production) low(er) ; 8 no ethical objections / no religious objections ; ora	4
3(b)(i)	$\frac{16}{32} \times 100$ or $\frac{100}{32} \times 16$; 50 ; A ecf to two or three significant figures	2
3(b)(ii)	any two from: 1 (bioinformatics is) a store / database, of, base / DNA / protein / amino acid / primary, sequence data ; 2 for comparison (of, database / base / DNA / protein / amino acid / primary, sequence data) ; 3 (to search for base / DNA / protein / amino acid / primary, sequences) similar to human calcitonin ; 4 <i>idea of</i> modelling / predicting, tertiary / 3D / protein, structure ; 5 AVP ; e.g. sequences data pooled from all over the world	2

Question	Answer	Marks
3(c)(i)	<u>restriction</u> , enzyme / endonuclease ; (DNA) <u>ligase</u> ;	2
3(c)(ii)	any two from: <i>one mark if only two features stated without explanation</i> 1 small so can be inserted (into cells) ; 2 replicate, independently / fast, so, high copy number / large number of plasmids or have origin of replication so high copy number / large number of plasmids ; 3 has restriction site(s) / can be cut by restriction enzymes, so (new) gene can be added ; 4 have, multiple cloning site / polylinker, so can be cut by different restriction enzymes ; 5 have marker genes so, recombinants / transformed bacteria , can be recognised / AW ; 6 circular so stable ;	2

Question	Answer	Marks
3(d)	<i>any two from:</i> 1 eukaryote and prokaryote promoter <u>sequences</u> are different ; 2 eukaryote and prokaryote RNA polymerase enzymes are different ; 3 prokaryotic RNA polymerase does not, recognise / bind to, eukaryotic promoter or prokaryotic RNA polymerase only binds to prokaryotic promoter ; 4 so no, transcription / mRNA made / gene expression ; ora 5 eukaryotic promoter requires binding of (many) transcription factors that are not present in prokaryotic cell ;	2